

Combinations

List all possible combinations.

1) T, V, W, taken two at a time

2) 1, 2, 3, 4, taken two at a time

3) 4, 5, 6, 7, taken three at a time

4) ☺, ☼, ♥, ▲, ■, taken two at a time

Evaluate each expression.

5) ${}_{18}C_{11}$

6) ${}_{24}C_4$

7) ${}_{17}C_5$

8) ${}_{18}C_5$

9) ${}_{22}C_4$

10) ${}_{15}C_9$

11) $-1 + {}_{23}C_{18}$

12) $1 + {}_{22}C_{18}$

13) $3 \cdot {}_{24}C_5$

14) ${}_{17}C_{12} + 2$

15) Explain why ${}_nC_3 = {}_nC_{n-3}$

16) Write a combination that equals 12345